

SYE6110: DATA ANALYSIS AND ARTIFICIAL INTELLIGENCE FOR SYSTEMS ENGINEERING

Effective Term

Semester A 2025/26

Part I Course Overview

Course Title

Data Analysis and Artificial Intelligence for Systems Engineering

Subject Code

SYE - Systems Engineering

Course Number

6110

Academic Unit

Systems Engineering (SYE)

College/School

College of Engineering (EG)

Course Duration

One Semester

Credit Units

3

Level

P5, P6 - Postgraduate Degree

Medium of Instruction

English

Medium of Assessment

English

Prerequisites

SYE4003 Artificial Intelligence and Advanced Technology in Manufacturing and Operations or SYE6106 Intelligent Manufacturing for Engineering Managers

Precursors

Nil

Equivalent Courses

ADSE6110 Data Analysis and Artificial Intelligence for Systems Engineering (offered until 2023/24)

Exclusive Courses

Nil

Part II Course Details

Abstract

As one of the most popular disciplines in recent years, data-driven Artificial Intelligence (AI), particularly machine learning-based approaches, is gradually transforming nearly all industries, such as domain of the engineering management. Advances in artificial intelligence, such as deep learning methods, are widely employed in many engineering industries for data analysis and research. Therefore, for those who wish to join the waves of AI, this course covers fundamental but comprehensive concepts and experiments of data analysis and machine learning, including but not limited to topics such as classification and regression, unsupervised clustering, machine learning, and an introduction to deep learning. This course provides hands-on opportunities to develop fundamental statistical, data-analysis, and AI models using the popular Python tools for smart manufacturing cases. This course will construct a solid foundation for more advanced courses in engineering, marketing, finance, economics, and more advanced data science, deep learning, and AI.

Course Intended Learning Outcomes (CILOs)

CILOs		Weighting (if app.)	DEC-A1	DEC-A2	DEC-A3
1	Identify the state-of-the-art key issues of a data analysis problem engineering management or manufacturing; and formulate these issues into machine learning or AI models for further analysis.	30	x	x	x
2	Apply the data analysis, machine learning knowledge acquired through the course to select the most appropriate technique for a given problem.	25		x	x
3	Analyze relevant data effectively using appropriate data analysis/machine learning techniques to solve the problems and evaluate the results in the context of the problems.	25		x	x
4	Develop the ability to use python module to conduct data analysis for manufacturing related problems.	20		x	x

A1: Attitude

Develop an attitude of discovery/innovation/creativity, as demonstrated by students possessing a strong sense of curiosity, asking questions actively, challenging assumptions or engaging in inquiry together with teachers.

A2: Ability

Develop the ability/skill needed to discover/innovate/create, as demonstrated by students possessing critical thinking skills to assess ideas, acquiring research skills, synthesizing knowledge across disciplines or applying academic knowledge to real-life problems.

A3: Accomplishments

Demonstrate accomplishment of discovery/innovation/creativity through producing /constructing creative works/new artefacts, effective solutions to real-life problems or new processes.

Learning and Teaching Activities (LTAs)

LTAs		Brief Description	CILO No.	Hours/week (if applicable)
1	Lecture	Concepts and specific subject knowledge are explained	1, 2, 3	13 hours/sem

2	In-class exercise	With the teacher acting as a facilitator, students work together on assigned engineering problem sets to consolidate their understanding of the concepts and methods. They are required to formulate the problem into a data driven model (define the machine learning inputs and targets) and proceed to solve the problem (the method). These exercises are based on data from real-life applications.	1, 2, 3, 4	26 hours/sem
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Assessment Tasks / Activities (ATs)

ATs		CILO No.	Weighting (%)	Remarks ("- " for nil entry)	Allow Use of GenAI?
1	Assignment or midterm	1, 2, 3, 4	50	-	No

Continuous Assessment (%)

50

Examination (%)

50

Examination Duration (Hours)

2

Minimum Continuous Assessment Passing Requirement (%)

30

Minimum Examination Passing Requirement (%)

30

Assessment Rubrics (AR)**Assessment Task**

1. Assignment (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

Criterion

Core concepts, ideas and coding of data driven software

Excellent

(A+, A, A-) High

Good

(B+, B, B-) Significant

Fair

(C+, C, C-) Moderate

Marginal

(D) Basic

Failure

F) Not even reaching marginal levels

Assessment Task

2. Examination (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

Criterion

Core concepts and ideas; use of appropriate data-driven, machine learning, or AI methods

Excellent

(A+, A, A-) High

Good

(B+, B, B-) Significant

Fair

(C+, C, C-) Moderate

Marginal

(D) Basic

Failure

(F) Not even reaching marginal levels

Assessment Task

1. Assignment (for students admitted from Semester A 2022/23 to Summer Term 2024)

Criterion

Core concepts, ideas and coding of data driven software

Excellent

(A+, A, A-) Excellent

Good

(B+, B) Good

Marginal

(B-, C+, C) Marginal

Failure

(F) Failure

Assessment Task

2. Examination (for students admitted from Semester A 2022/23 to Summer Term 2024)

Criterion

Core concepts and ideas; use of appropriate data-driven, machine learning or AI methods

Excellent

(A+, A, A-) Excellent

Good

(B+, B) Good

Marginal

(B-, C+, C) Marginal

Failure

(F) Failure

Part III Other Information

Keyword Syllabus

- Introduction to AI, machine learning, deep learning, and their applications in engineering and engineering management
- Python programming
- Linear regression with multiple approaches
- Bayes' Theorem
- Unsupervised learning and kmeans
- Logistic regression
- Decision tree
- Support vector machine
- Deep learning and neural networks
- Case studies for engineering and manufacturing applications

Reading List

Compulsory Readings

Title	
1	Nil

Additional Readings

Title	
1	Python Crash Course, 2nd Edition: A Hands-On, Project-Based Introduction to Programming 2nd Edition
2	Statistics for Business: Decision Making and Analysis, by Robert Stine and Dean Foster
3	Deep Learning with Python
4	Machine Learning, Andrew Ng, coursera.org